ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 D ✓
1.2 B ✓
1.3 C ✓
1.4 A ✓
1.5 B ✓
1.6 D ✓
1.7 A ✓
1.8 C ✓
1.9 C ✓
1.10 A ✓

(10 × 1) [10]

QUESTION 2

2.1 False ✓
2.2 True ✓
2.3 True ✓
2.4 True ✓
2.5 True ✓
2.6 False ✓
2.7 False ✓
2.8 True ✓
2.9 True ✓
2.10 False ✓

(10 × 1) [10]

QUESTION 3

3.1 F ✓
3.2 C ✓
3.3 J ✓
3.4 E ✓
3.5 H ✓
3.6 D ✓

3.7 G ✓

3.8 A ✓

3.9 I ✓

3.10 B ✓

(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 CPU ✓; flash (program) memory ✓; RAM ✓; I/O ports ✓; timers ✓ and ADC ✓, all connected by an internal bus (address/data/control) (6)

4.2 A microcontroller has the CPU, memory and I/O on one chip, designed to control one dedicated task ✓, e.g. ATmega328P (Arduino Uno), PIC, ESP32 ✓; a microprocessor is only the CPU and needs external memory and I/O, used in general-purpose computers ✓, e.g. Intel Core i5, the processor of a Raspberry Pi / ARM Cortex-A ✓. (4)

4.3 UART: 2 wires (TX, RX) ✓; slow to medium (e.g. 9 600–115 200 baud) ✓; GPS, Bluetooth module, PC serial monitor ✓. I²C: 2 wires (SDA, SCL) ✓; medium (100–400 kHz) ✓; IMU, OLED display, real-time clock ✓. SPI: 4 wires (MOSI, MISO, SCK, SS) ✓; fast (several MHz) ✓; SD card, TFT display, radio module ✓ (9)

4.4 With polling, the program must keep checking the pin and can miss pulses while it is busy ✓; an interrupt responds immediately to every pulse ✓, so counting is accurate and the main program can do other work ✓. (3)

[22]

QUESTION 5

5.1.1 $A_v = 3{,}3 \div 0{,}3 \checkmark = 11 \checkmark$ (2)

5.1.2 $A_v = 1 + R_f/R_i \rightarrow 11 = 1 + R_f/1\ \text{k}\Omega \checkmark; R_f = 10 \times 1\ \text{k}\Omega \checkmark = 10\ \text{k}\Omega \checkmark$ (3)

5.1.3 Op-amp triangle ✓; input to the non-inverting (+) terminal ✓; R_f from output to the inverting (−) terminal ✓; R_i from the inverting terminal to 0 V ✓ (4)

5.2 $V_{out} = -(R_f/R_i) \times V_{in} \checkmark = -(100/20) \times 0{,}5 \checkmark = -2{,}5\ \text{V} \checkmark$ (3)

5.3 $1000\ 0000 = 128 \checkmark; V = 128 \times 5 \div 256 \checkmark = 2{,}5\ \text{V} \checkmark$ (3)

[15]

QUESTION 6

6.1 Diagram: four switches S1–S4 arranged in an H, with the motor across the middle ✓✓, supply at the top and 0 V at the bottom ✓. Forward: S1 (top left) and S4 (bottom right) closed ✓✓; reverse: S2 (top right) and S3 (bottom left) closed ✓✓ (accept any consistent labelling) (7)

6.2 It would short-circuit the supply (shoot-through) ✓, causing a very high current that can destroy the transistors or battery ✓; driver ICs have built-in logic / dead time that prevents both switches on one side from being on together ✓. (3)

6.3 When the motor stalls or starts, it can draw more current than the driver can supply, so the driver may overheat or be damaged ✓. Any TWO ✓✓: use a driver rated above the stall current (e.g. 3 A or more); add current limiting / fuse; use a heat sink and cooling; avoid stalling with gearing; use soft starting with PWM ramps (3)

[13]

QUESTION 7

7.1.1 $\mathrm{P} = (\mathrm{V_{in}} - \mathrm{V_{out}}) \times \mathrm{I}$ ✓ $= (9 - 5) \times 0{,}5$ ✓ $= 2\ \mathrm{W}$ ✓ (3)

7.1.2 2 W makes the regulator very hot, and it may shut down or be damaged ✓; a heat sink spreads the heat ✓; a switching (buck) converter is more efficient (about 85–95%) ✓. (3)

7.2 $\mathrm{t} = \text{capacity} \div \text{current} = 2\,200 \div 550$ ✓ $= 4\ \mathrm{h}$ ✓; reasons (any TWO) ✓✓: current peaks when motors start or climb; battery should not be fully discharged; capacity drops with age and temperature; voltage sag (4)

7.3 Any FIVE ✓✓✓✓✓: use a proper LiPo balance charger; never leave batteries charging unattended; charge on a fire-proof surface or in a LiPo bag; do not charge damaged or swollen batteries; do not over-discharge (stop at about 3,0–3,3 V per cell); store at storage voltage (about 3,8 V per cell) in a cool place; avoid short circuits / cover terminals (5)

7.4 A common ground gives all signals the same 0 V reference ✓, so the microcontroller's control signals are read correctly by the motor driver ✓. Large, changing motor currents cause voltage drops and electrical noise ✓, which can reset the microcontroller or corrupt sensor readings ✓; so the grounds are joined at one point (star ground) and motor wiring is kept separate / decoupling capacitors are used ✓. (5)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE